

# MATTHEW H. ZHANG

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## EDUCATION

The University of Texas at Austin – GPA: 3.73

May 2028

*Bachelor of Science in Electrical and Computer Engineering Honors*

*Bachelor of Business Administration in Canfield Business Honors*

**Relevant Coursework:** Circuit Analysis & Design Honors, Computing Systems Honors, Embedded Systems Honors, Software Design & Implementation Honors, Management Information Systems Honors

## EXPERIENCE

**Venture Analyst Intern, Humanoid Teleoperation Collaboration – Tess Ventures**

Mar 2026 - Present

- Led development of a wearable rig capturing human arm joint angles at 50 Hz via custom binary packet
- Transmitted motion data over **USB serial** to teleoperate a **Unitree G1 EDU** humanoid
- Prepared **technical demos and product-facing presentations** for engineering and business audiences

**Control Systems & Electrical Engineer – Texas Aerial Robotics**

Jan 2026 - Present

- Modeled inverted pendulum and TVC system dynamics and implemented **PID, MPC, & LQR** controllers in Python
- Designing a compact AM32-based **30A 4S BLDC motor driver PCB** under size and thermal constraints in Altium

**Embedded Systems Engineer – GHOST Robotics**

Nov 2025 - Present

- Validated custom sensor PCBs and integrated them into tight mechanical assemblies using Onshape
- Performed board bring-up and debug by validating power rails and signals with a multimeter and logic analyzer
- Wrote interrupt-based **RP2040 sensor polling firmware** in C with custom sensor drivers over USB serial

**Team Captain, Lead Engineer – BagelBytes Robotics**

Dec 2024 - Aug 2025

- Organized a team of **30+** individuals to successfully design, manufacture, and field functioning 125 lb robots
- Mentored **20+** members in CAD & electrical systems; **improved team ranking by 1,500+ places** during leadership
- Engineered custom **CAN bus** layout and architecture, reducing match-time electrical failures by **28%**
- Designed **100+ custom sheet metal parts** and 7 custom-built gearboxes with Onshape

## PROJECTS

**Personal Project Vinyl Record Player**

Dec 2025 - Present

- Independently designed MCU-based motor driver 2-layer PCB with **USB-C PD** power negotiation for turntable
- Created power distribution with custom **buck converter** (12V → 5V), local decoupling, differential-pair routing

**1st Place, \$6000 Prize - UT Genesis x Momentum x Yconic Hackathon**

April 2026

- Built an ESP32 **wearable AI assistant** with face recognition, live STT, and persistent memory of conversations
- Developed a pipeline for face tracking, identity matching, and context retrieval across repeated interactions

**1st Place - IEEE Robotics & Automation Society, Robotathon Competition**

Sep 2025 - Nov 2025

- Led electromechanical design of autonomous competition robot that launched balls with custom flywheels
- Adjusted designs for modularity, allowing for easy sensor iteration and improved maze routing

**STARDUST Wall Projector Pen - HackTX**

Oct 2025

- Led team of 4 under tight time constraints during a 48-hour hackathon to develop a pocket-sized whiteboard
- Embedded sensors and **BLE** firmware with **OpenCV** to map pen pressure and coordinates to writing on the wall

## SKILLS

**Technical Skills:** Circuit Design, PCB Layout, Firmware Development, CAD, Git, Debugging, Control Systems, Soldering

**Tools:** KiCad, Altium, LTSpice, MuJoCo, Linux, Onshape, GDB, Oscilloscope, Logic Analyzer, Excel, Machining & Shop Tools

**Languages:** C, C++, Python, Assembly, R, SQL | **Embedded:** STM32Cube, ESP-IDF, CCSTUDIO, I2C, SPI, UART, BLE

Yconic Hackathon 1st Place - \$6000 (2026), IEEE RAS Robotathon 1st Place, Software Design Award (2025), FIRST Robotics Rising All Stars Award (2025), World's Rookie Inspiration Award, FIRST Robotics Worlds Qualifier (2024)